VU Medical Zone

BT 503 Final Term Subjective

Q1. Define Halophiles. **2 Marks**

Halophiles survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds.

Q2. Define psychrophile. **2 Marks**

Cold environments are more common on earth than hot ones. Average oceanic temperature is around 1–3°C and vast areas of the global land mass are permanently or near permanently frozen. Psychrophiles can tolerate these circumstances.

Q3. Define Biosensor. **2 Marks**

It is as analytical devices incorporating a biological material, a biologically derived material, or biomimic, intimately associated with or integrated within a physicochemical transducer or transducing microsystem.

Q4. Define Co-metabolism. **2 Marks**

Cometabolism: a xenobiot is degraded by the organism's enzymes but in this case its catabolism does not provide energy and so cannot be the sole carbon source.

Q5. How microbes are poly cyclic aromatic compound? **2 Marks**

Bacteria against PAHs (polycyclic aromatic hydrocarbons, e.g., naphthalene and phenanthrene). Bacteria isolated from the same environments may vary in their abilities to degrade PAHs indicative of diverse catabolic pathways.

Q6. Write two advantages of the conventional biological sludge dewatering process. **2 Marks**

i. Ease to installation.

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ii. Low land requirement.

iii. Operation under high loading rate.

Q7. Mention two Natural sludge dewatering process. **2 Marks**

Natural sludge dewatering processes:

• Drying beds

• Sludge lagoons

Q8. How quality of fruit should improve? **2 Marks**

The quality of the fruit was maintained by virtue of the fact that the sodium accumulation caused by the antiport occurred in leaves only and not in the fruit.

Q9. Write two sources of dioxins? **2 Marks**

Dioxins released as byproducts in a number of chemical processes.

 Production of some pesticides, the manufacture of PVC plastics, the chlorine bleaching of pulp and paper, and waste incineration.

 Carcinogenic.

Q10. What are xenobiotics? **2 Marks**

Greek word 'xenos' meaning foreign. Xenobiotics are compounds which are man-made and not produced by a biological procedure and for which no equivalent exists in nature.

Q11. Ex-situ technique? **2 Marks**

The main characteristic of ex situ methods is that the soil is removed from where it originally lay, for treatment. These techniques are best suited to instances of relatively localized pollution within a site, typically in 'hot-spots' of medium to relatively high concentration which are fairly near to the surface.

Q12. What is Glyphosate? **2 Marks**

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Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5-enolpyruvylshikimate-3- phosphate synthase. The gene coding for this enzyme has been identified, isolated and inserted into a number of plants e.g., Petunias.

Q13. Define bioscrubber. **2 Marks**

The bioscrubber is not itself truly a biological treatment system, but rather a highly efficient method of removing odour components by dissolving them.

Q14. What is sustainable bioremediation? **2 Marks**

Sustainable Remediation Forum (SuRF) UK framework document identified six 'key principles of sustainable remediation', namely:

1. Protection of human health and the wider environment.

2. Safe working practices.

3. Consistent, clear and reproducible evidence-based decision making.

4. Record keeping and transparent reporting.

5. Good governance and stakeholder involvement.

6. Sound science.

Q15. Why DDT is banned? **2 Marks**

DDT causes weakening of egg shells of birds, which led to dramatic decreases in the populations of many species.

Q16. Consequence of OECD 2001 about global biotechnology? **2 marks**

The OECD (2001) concluded that the industrial use of biotechnology commonly leads to increasingly environmentally harmonious processes and additionally results in lowered operating and/or capital costs.

Q17. Define Phytodegradation and name its forms. **2 Marks**

Phytodegradation Biological breakdown of contaminants by plants.

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Two forms

• Internally, having first been taken up by the plants.

• Externally, using enzymes secreted by them.

Q18. What are living conditions of extremophiles? Example. 2 marks

Some microbial species tolerate extreme conditions, like high salinity, pressures and temperatures for example halophile.

Q19. Difference between bioenhancement and biogumentation. 2 Marks

Bioenhancement concentrates solely on the existing microfauna, stimulating their activity by the manipulation of local environmental conditions.

Bioaugmentation requires the deliberate introduction of selected microbes to bring about the required clean-up.

Q20. Difference between natural and conventional sludge dewatering process. 3 Marks

Natural sludge dewatering processes:

• Drying beds

• Sludge lagoons

Conventional sludge dewatering processes:

The biological stage comprises two units: aeration tank (reactor) and secondary sedimentation tank. High biomass concentration in the reactor due to the recirculation of the settled bacteria from the bottom of the secondary sedimentation tank.

Q21. Name the enzyme, solution and equipment used in genetic engineering. 3 Marks

The enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell.

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Q22. There classes of biosensor and two advantages and disadvantages of immunosensor. **3 Marks**

 According to the biorecognition principle, biosensors classified into

• immunochemical,

• enzymatic,

• nonenzymatic receptor,

• whole-cell, and

• DNA biosensors.

 Immunosensors present the advantages of sensitivity and selectivity.

 Limitations are the regeneration of the immune surface, and cross reactivity.

Q23. Enlist the type of solid waste. **3 Marks**

Organic waste: Waste from preparation of food, market places, etc.

Combustibles: Paper, wood, dried leaves, packaging, etc. (high organic and low moisture content)

Non-combustibles: Metal, tin cans, bottles, stones, etc.

Ashes/dust: Residue from fires used for cooking.

Bulky waste: Tree branches, tyres, etc.

Dead animals: Carcasses of domestic animals and livestock.

Hazardous waste: Oil, battery acid, medical waste.

Construction waste: Roofing, rubble, broken concrete, etc.

Q24. What is phytoplankton? **3 Marks**

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Phytoplanktons, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop. Most are light and swim in the upper portion of the sea, where light infiltrates the water.

Q25. How quality of crops should enhance? 3 Marks

Crop quality improvements:

Control of fruit ripening, production of cereals with improved nutritional value. These although are of great interest to the food industry, they are of more peripheral relevance to environmental biotechnology.

Q26. Write 3 oxidation product of TNT. 3 Marks

• 4-amino-2,6- dinitrotoluene (4ADNT)m

• 2-amino-4,6- dinitrotoluene (2ADNT)

• 2,4-diamino-6-nitrotoluene (2,4DANT)

Q27. Process of Thickening of sludge. 3 Marks

The main processes used for sludge thickening are:

• gravity thickeners

• dissolved air flotation

• centrifuges

• belt presses

Q28. Write features of phenols that makes it pollutant. 3 Marks

Phenol is a weak acid (pKa = 9.82) and thus not ionized at pH 7.

Being ionizable, the phenols are quite soluble in water and are commonly found as groundwater contaminants.

Q29. What is T-DNA? 3 Marks

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Wild type plasmid contains genes which causes the transfer of a piece of DNA, 'T-DNA', into a plant cell, causing crown gall disease. This piece is flanked by 24 bp direct repeats. These genes may be cut out and replaced by DNA containing the gene of choice to be introduced into the plant.

Q30. What is Environmental Biotechnology? 3 Marks

Environmental biotechnology is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.

Q31. Biological remediation. 3 Marks

Biological: This involves the transformation or mineralisation of contaminants to less toxic and more mobile, forms. This can include fixation or accumulation in harvestable biomass crops.

Biological: Advantages:

• can destroy a wide range of organic compounds,

• can cause benefit to soil structure and fertility,

• are non-toxic.

Disadvantages:

• the process end-point can be uncertain and difficult to gauge,

• treatment can be slow

• not all contaminants are conducive to treatment by biological means.

Q32. Advantages of association of fungi with plant? 3 Marks

Mycorrhizae: Associations of fungi with roots of vascular plants Quite common and in some cases be very beneficial to the plant.

They may be

• external, ectomycorrhizal,

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• internal, endomycorrhizal.

Q33. Describe the importance of anaerobic zone in biological process of phosphorus removal in activated sludge treatment? 3 Marks

It is essential to have anaerobic and aerobic zones in the treatment line for the biological removal of phosphorus. The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms. For higher efficiencies of phosphorus removal, effluent polishing methods can adopt: Addition of coagulants (metallic ions): phosphorus precipitation • effluent filtration: removal of the phosphorus present in the suspended solids in the effluent • combination of the addition of coagulants and filtration.

Q34. Describe the natural process to remove the pathogenic environment. 5 Marks

Maturation ponds: Shallow ponds, where the penetration of solar UV radiation and unfavourable environmental conditions causes a high mortality of the pathogens. The maturation ponds do not need chemical products or energy, but require large areas. They are highly recommended systems.

Land treatment (infiltration in soil): The unfavourable environmental conditions in the soil favour the mortality of the pathogens. Chemical products are not needed. Requires large areas.

Q35. Environmental concern to the release of the hormone into environment. 5 Marks

Many environmental contaminants produce adverse effects by interfering with endogenous hormone systems, the so-called EDCs. EDCs constitute a class of substances not defined by chemical nature, but by biological effect. “Endocrine effect biosensors” take advantage of this feature. Endocrine Effect Biosensors. Steroid hormones induce different effects in mammalian cells after binding to specific intercellular receptors (ligand-dependent transcription factors). Many endocrine disruptors also believed to bind to the estrogen receptor (ER) as agonists or antagonists. Binding ability of the chemicals toward the ER crucial for screening or testing their

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potential environmental toxicity. Several biosensors are based on ER. Simple to perform and allow identification of all endocrine disruptors that act through the corresponding receptor.

Q36. Role of filter medium in boiler also name the compound of the filter medium. **5 Marks**

The first methods to be developed. Consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic, which holds a filter medium of organic material such as peat, heather, bark chips and the like. The gas to be treated is forced, or drawn, through the filter. The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. Bacteria, and other micro-organisms present, degrade components of the resultant solution, thereby bringing about the desired effect. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity. Biofilters need to be watered sufficiently to maintain optimum internal conditions, but waterlogging is to be avoided as this leads to compaction, and hence, reduced efficiency. Properly maintained, biofilters can reduce odour release by 95% or more.

Q37. Five impacts of chlorobenzene & dioxin on human health. **5 Marks**

Human responses include chloracne, weight loss, insomnia, liver dysfunction, spontaneous abortion, and possibly cancer.

Q38. Write a note on cDNA libraries. **5 Marks**

cDNA libraries: In eukaryotes, the first product of transcription from DNA is not messenger RNA (mRNA) but heterogeneous nuclear RNA (hnRNA). This is mRNA prior to the removal of all the non-coding sections, or introns, which are discarded during the processing to produce the mature mRNA. DNA for transfer cDNA libraries cDNA is DNA which has been artificially made using the mature mRNA as a template (through Reverse Transcriptase enzyme). It is then used as the template for the second strand. Thus, the synthetic DNA product is simply a DNA version of the mRNA and so should overcome the problems of expression mentioned ago.

Q39. Write five properties of Soil. **5 Marks**

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In case of soil particularly, properties such as

• texture,

• porosity,

• humus content,

• moisture

• microbial complement

Q40. Write five advantages of Biofilter. **5 Marks**

The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity.

Q41. Explain vitrification method. **5 Marks**

Solidification / vitrification: It is the encapsulation of contaminants within a monolithic solid of high structural integrity, with or without associated chemical fixation. Vitrification uses high temperatures to fuse contaminated materials.

Solidification / vitrification:

Advantage: Toxic elements and/or compounds which cannot be destroyed, are rendered unavailable to the environment.

Disadvantages: • Contaminants are not actually destroyed. • Significant amounts of reagents are required and not suitable for organic contaminants.

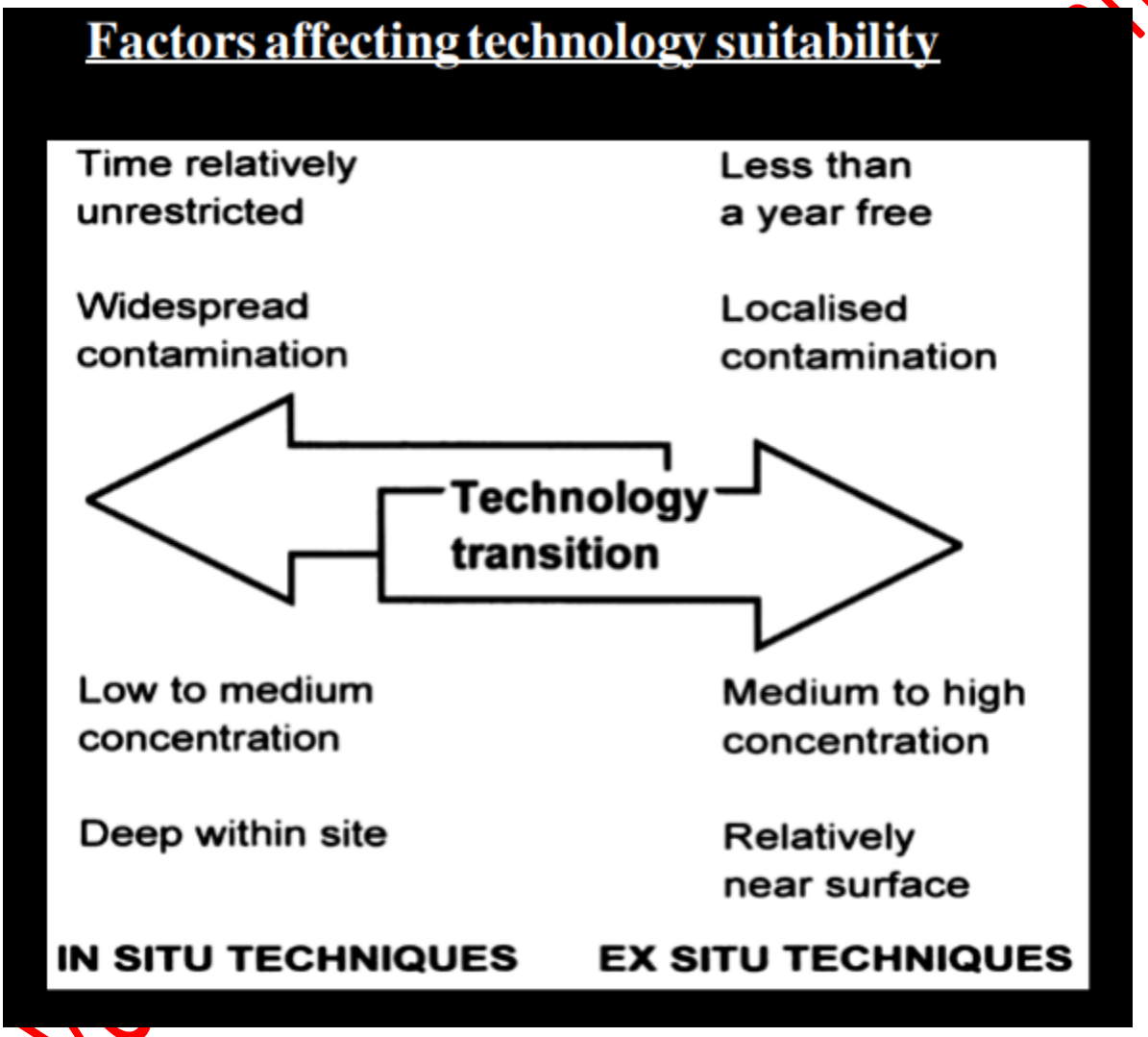
Q42. Genetic modification a popular plant is used for removal of mercury. Describe How? **5 Marks**

Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere. mer A gene gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

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Initially the gene was transferred to Arabidopsis thaliana. The gene was transferred by gene guns to poplar tree (Liriodendron tulipifera) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatalise it at 10 times the rate observed in untransformed yellow poplar plantlets.

Q43. Factors affecting the technique in situ and ex situ. **5 Marks**



Q44. Application of biosensor for the detection of biocide? **5 Marks**

Widespread presence of pesticides in natural waters. Concerns about their toxicity and persistence led the European Community to set limits on the concentration. A limit of 0.1 µg/l for individual pesticides and of 0.5 µg/l for total pesticides. Though techniques such HPLC/MS and GC/MS give satisfactory analytical results for pesticide determination. New assays and

sensors for cheaper and faster onsite analysis. Enzymatic sensors, based on the inhibition of a selected enzyme, are the most extended biosensors used for the determination of these compounds. Various biosensors, based on the inhibition of acetyl cholinesterase (AChE) and colin oxidase, for the detection of organophosphorous and carbamate pesticides. Although sensitive, biosensors based on AchE inhibition are not selective. One approach to solve the lack of specificity of AchE involves the genetic engineering of cholinesterase enzyme.

Q45. Explain Genomic libraries. 5 Marks

Genomic libraries Collection of total Genomic DNA from an organism. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell. These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain non-coding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

Q46. Solid waste step? 5 Marks

1. Identify the types of waste.

2. Identify the sources of waste.

3. Determine the potential health hazards from waste.

4. Determine the volume of waste generated

5. Identify safe collection method/s.

6. Identify safe transportation method/s.

7. Identify safe disposal method/s.

Q47. Write down the source of solid waste? 5 marks

Sources of solid waste:

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• Medical centers • Food stores • Feeding centers • Food distribution points • Slaughter areas • Warehouses • Markets • Domestic areas

Sources of solid waste: Solid waste management strategies may vary for institutional, communal and domestic sources, and Depends on the types and volumes of the waste

Q48. What are the environmental factors that affect the bioremediation describe briefly? 5 Marks

Environmental Factors: Most significant are:

Temperature: 0 and 50 ◦C. Most efficient: 20–30 ◦C

pH: pH range of 5.0-9.0, however, 6.0-8.0 being most efficient.

Soil type: sands and gravels are the most suitable soil types for bioremediation

Other factors:

• Nutrient availability, • oxygenation, • presence of other inhibitory contaminants • whether the site is contained or if the groundwater runs off, • what contaminants are present, what is their concentration and whether they are biodegradable, etc.

Q49. What is PCBS? Which biosensors detect them? 5 Marks

Polychlorinated biphenyls (PCBs) are xenobiotics having high level of halogenation. PCBs are substrates for very few pathways normally occurring in nature.

DNA biosensor with chronopotentiometry detection, and various immunosensors with fluorescence, SPR, and electrochemical detection principles.

Q50. Mention some metallic coagulant for sludge dewatering? 5 Marks

The most common metallic coagulants are:

• Aluminium sulphate

• Ferric chloride

• Ferrous sulphate

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• Ferric sulphate

• Quicklime/ Hydrated lime

Q51. Explain Phytovolatilization. 10 marks

Phytovolatilization Involves the uptake of the contaminants by plants and their release into the atmosphere, typically in a modified form. Relies on the transpiration pull of fast growing trees, which accelerates the uptake of the pollutants in ground water solution, which are then transformed and released through the leaves. E.g., genetically modified variety of the Yellow Poplar, Liriodendron tulipifera Engineered by the introduction of mercuric reductase gene (mer A). This confers the ability to tolerate higher mercury concentrations, and removal of the pollutant from the soil and volatize it. In a number of studies, poplars have been shown to be able to volatilize around 90% of the TCE (Trichloroethylene) they take up. Phytoremediation tends to be limited to sites where the pollutants are located fairly close to the surface. However, recent research has shown that deeply penetrating roots of trees allow deeper contamination to be treated.

Q52. Dental not sludge treatment stage before final disposal steps? 10 Marks

Main final disposal alternatives for the sludge:

• Ocean disposal,

• Incineration,

• Sanitary landfill,

• Landfarming,

• Land reclamation,

• Agricultural reuse.

Ocean disposal: After pre-conditioning, the sewage is disposed in the sea, through ocean outfalls or barges. Disposal without beneficial uses.

Ocean disposal:

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Advantages: • Low cost

Disadvantages: • Ocean water, flora and fauna pollution.

Incineration: Thermal decomposition process by oxidation, in which the volatile solids of the sludge are burnt in the presence of oxygen and are converted into carbon dioxide and water. The fixed solids are transformed into ashes.

Incineration:

Advantages: • Drastic volume reduction, • Sterilisation.

Disadvantages: • High costs, • Ash disposal, • Atmospheric pollution.

Sanitary landfill: Disposal of the sludges in ditches or trenches, with compaction and covering with soil, after which they are sealed. (can be co-disposed with urban solid wastes). Disposal without beneficial uses.

Sanitary landfill:

Advantages: • Low cost

Disadvantages: • Requirement of large areas, • Problems with locations near urban centres, • Requirement of special soil characteristics, • Gas and leachate production, • Difficulty in reintegrating the area after decommissioning.

Landfarming: Land disposal process, in which the organic substrate is biologically degraded in the upper layer of the soil and the inorganic fraction is transformed or fixed into this layer. Disposal without beneficial uses.

Landfarming:

Advantages: • Low cost • Disposal of large volumes per unit area

Disadvantages: • Accumulation hardly decaying constituents…, • Possible groundwater contamination, • Odour release and vector attraction, • Difficulty in reintegrating the area after decommissioning

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Land reclamation: Disposal of sludge in areas that have been drastically altered, such as mining areas, where the soil does not offer conditions for development and fixation of vegetation, as a result of the lack of organic matter and nutrients.

Land reclamation:

Advantages: • High application rates, • Positive results for the recovery of the soil and flora. Disadvantages: • Odours, • Composition and use limitations, • Contamination of the groundwater, fauna and flora.

Agricultural reuse: Disposal of the sludge in agricultural soils, in association with the development of crops. Beneficial use of the sludge (which, in this case, is named as a biosolid).

Agricultural reuse:

Advantages: • Large area availability, • Positive effects on soil, • Long term solution, • Potential as a fertilizer, • Positive outcome for crops.

Disadvantages: • Limitations regarding composition and application, • Contamination of the soil by metals, • Food contamination with toxic elements pathogens • Odours.

Q53. Describe biosensor single transduction and bio recognition principle? 10 Marks

Classified according to the signal transduction and the biorecognition principles.

On the basis of the transducing element, categorized as electrochemical, optical, piezoelectric, and thermal sensors.

Electrochemical biosensors, and among them the amperometric and the potentiometric ones, are the best described in the literature. Those based on optical principles are the next most commonly used transducers.

According to the biorecognition principle, biosensors classified into • immunochemical, • enzymatic, • nonenzymatic receptor, • whole-cell, and • DNA biosensors. ¬ Immunosensors present the advantages of sensitivity and selectivity. ¬ Limitations are the regeneration of the immune surface, and cross reactivity.

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Q54. Write down the procedure of sludge treatment? 10 marks

The main stages in sludge management:

• Thickening: removal of water (volume reduction)

• Stabilisation: removal of organic matter (mass reduction)

• Conditioning: preparation for dewatering

• Dewatering: removal of water (volume reduction)

• Disinfection: removal of pathogenic organisms

• Final disposal: final destination

Thickening: It is a physical process of concentrating the sludge, with the aim of reducing its water content and, as a result, its volume, facilitating the subsequent sludge treatment stages.

Stabilisation: Stabilisation aims at attenuating the inconveniences associated with the generation of bad odours and disposing of the sludge. This is accomplished through the removal of the biodegradable organic matter of the sludge.

Conditioning: Conditioning is a sludge preparation process, based on the addition of chemical products (coagulants, polyelectrolytes) to increase its dewatering capability and to improve the capture of solids in the sludge dewatering systems.

Dewatering: It can be done through natural or mechanical methods. The objective of this phase is to remove water and reduce the volume even further, producing a sludge with a mechanical behaviour close to solids. This has an important impact in its transport and final disposal costs.

Disinfection: This step is necessary, since the anaerobic or aerobic digestion processes usually employed do not reduce the pathogens content to acceptable levels. Disinfection is not necessary if the sludge is to be incinerated or disposed of in landfills.

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Q55. Explain the Southern and western blotting. 10 Marks

The next stage is normally to analyze the DNA isolated from possible recombinants, firstly by checking the size of the molecule or pieces thereof, or by sequencing the DNA. However, if a

large number of samples are to be analyzed it is usually quicker and cheaper to scan them by a procedure described as a Southern blot.

In Southern blotting, DNA fragments are resolved by electrophoresis on a gel. These resolved fragments are then shifted to a membrane (nylon or nitrocellulose). The membrane is then probed by a piece of DNA complementary to the sequence of interest. If a band shows up on autoradiography then the probe has found a mate and the required sequences are present. DNA sequencing is then required to confirm the results.

Western blotting: Proteins (e.g., antibodies antigens interaction)

Q56. Cloning Vector. 10 Marks

A cloning vector is frequently a plasmid or a bacteriophage (bacterial virus) Must be fairly small and fully sequenced, able to replicate itself when reintroduced into a host cell, thus producing large amounts of the recombinant DNA for further manipulation. Also it must carry on it 'selector marker' genes, which are indicators of genomic integrity and activity. A common design of a cloning vector is one which carries two genes coding for antibiotic resistance. The 'foreign' gene is inserted within one of the genes so that it is no longer functional therefore it is possible to discriminate by standard microbiology techniques, which bacteria are carrying plasmids containing recombinant DNA and which are not. To make the manipulations easier, selector marker genes normally contain a multi cloning site (MCS) which is a cluster of sites for restriction enzymes. Insertion of foreign DNA in vector at MSC site causes disruption of these marker genes resulting in loss of the function of that gene. For example, if that gene codes for antibiotic resistance, it will no longer be functional and will not protect the bacterium from that antibiotic. Viruses may be used as vectors i.e., the DNA of viruses can be used to carry foreign DNA. Most such recombinant viral DNA vectors need phage particles to perform proper gene transfer i.e. transduction. Transfer of naked viral recombinant vector to host cells is called transfection, and tends to have lower uptake rates as compared to transduction.

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Q57. Briefly describe the categories of solid waste? 10 Marks

Different categories of solid waste include:

Organic waste: Waste from preparation of food, market places, etc.

Combustibles: Paper, wood, dried leaves, packaging, etc. (high organic and low moisture content)

Non-combustibles: Metal, tin cans, bottles, stones, etc.

Ashes/dust: Residue from fires used for cooking.

Bulky waste: Tree branches, tyres, etc.

Dead animals: Carcasses of domestic animals and livestock.

Hazardous waste: Oil, battery acid, medical waste.

Construction waste: Roofing, rubble, broken concrete, etc.

Q58. What are the risks of solid wastes regarding transfer of disease? 10 Marks

Associated risks:

Disease transmission Decomposing organic waste attracts animals, vermin and flies. Flies may play a major role in the transmission of faecal-oral diseases. Rodents may increase the transmission of diseases such as Leptospirosis. Solid waste may also provide breeding sites for mosquitoes… Aedes genus… Dengue fever, Anopheles genus… Malaria. In times of food scarcity, members of the affected population may be attracted to waste heaps to scavenge for food. Likely to increase the risk of gastro-enteritis, dysentery and other illnesses.

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